ANDREY OSIPOV

Python Backend Engineer

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GENERAL INFORMATION

Employment: Full-time
Work format: Remote, hybrid, on-site
Location: Belgrade, Serbia
Citizenship: Russia, Serbian residence permit

SUMMARY

Python backend engineer with 3+ years of commercial experience in high-load services on FastAPI, Django and Flask: performance optimization, large-scale data, testing, observability, CI/CD and Kubernetes deployments. Own services end to end – from database schema and API design to production and incidents. Shipped ML and LLM features: a neural-network photo deduplication service and LLM-assisted user support. Degree in information security – I write code with security and load in mind.

EXPERIENCE

Total commercial experience – 3+ years.

Safe Transport Innovation Center

Sep 2025 – Present

Python Developer — *Remote*

Services for Moscow transport-security systems: main service on Flask, newer services on FastAPI. Sensitive data under NDA.

- Designed an aggregated data model over a multi-terabyte MongoDB store (**8.6B records**) from dozens of collections, cutting full object summary retrieval **from 3.2s to 87ms** and database load **by 37%**.
- Built a **neural-network photo deduplication service** (FastAPI, PyTorch): MobileNetV3 embeddings, YOLO11 and CNN detectors, grouping via graph connected components; **up to 97% accuracy** on **up to 10,000 photos a day**, explainable matches, **122 tests**.
- Built from scratch a **Git-style state-versioning system** for the core service in about two weeks with Claude Code as a coding agent; regressions between releases became traceable and acceptance testing got **1.7x** faster.
- Migrated inherited Flask services, including the deduplication service, to async FastAPI with API contracts intact: p95 latency **from 480ms to 85ms**, throughput **from 450 to 720 RPS**.
- Introduced **Sentry** and Telegram uptime alerting across my services, plus health checks with hang detection and auto-rollback deploys for the deduplication service; time to detect incidents fell **from 4 hours to 5 minutes**.
- Set up **OAuth 2.0 via Authentik** and JWT auth for my services; input validation, access control and dependency review let services with sensitive data **pass external security audits**.

Stack: Python, FastAPI, Flask, PyTorch, MongoDB, Redis, ClickHouse, RabbitMQ, Docker, Authentik, Sentry, asyncio, threading, multiprocessing, mypy, pyright, Claude Code

Rutube

Sep 2024 – Aug 2025

Python Developer — *Moscow*

Leading Russian video platform. Django (DRF) monolith plus aiohttp, FastAPI and Go microservices communicating over Kafka, heavy processing offloaded to Celery; team of 13 engineers; owned my services end to end – from development and code review to running them in production on Kubernetes.

- Optimized **10+ high-load API endpoints** using Prometheus and Grafana data: reworked ORM queries (select_related/prefetch, indexes), added Redis caching and rate limiting and cut p95 latency **from 1.1s to 92ms** at **~16,000 RPS**.
- Rewrote heavy SQL queries using Django CTEs and window functions, cut their execution time **from 12 minutes to 8 seconds** and removed critical load from PostgreSQL.

- Moved **200+ tests** to pytest with parallel CI runs (pytest-xdist) and raised coverage **from 42% to 78%**.
- Migrated API docs to OpenAPI 3.0 (drf-spectacular), documented **157 endpoints** and gave neighboring teams clear API contracts.
- Ran a security audit of the services, closed **OWASP Top 10 vulnerabilities**, updated insecure dependencies and tightened access policies.
- Unified Dockerfiles for two services and adopted the uv package manager, cutting CI image builds **from 8.5 to 2 minutes**; deployed and operated services on **Kubernetes via Rancher**: rollouts, pod management and incident investigation inside pods.

Stack: Python, Django, DRF, aiohttp, FastAPI, PostgreSQL, Redis, Kafka, Celery, gRPC, Cassandra, Elasticsearch, Prometheus, Grafana, Docker, Kubernetes (Rancher), pytest, microservices

PSK SOFT LLC

May 2023 – Aug 2024

Python Developer

Moscow

Backend development of corporate web services on Python/Django with PostgreSQL: server-side logic, integrations and production infrastructure.

- Built the **backend of 4 corporate web projects** on Django (DRF) and PostgreSQL: domain models, business logic and admin panels.
- Implemented server-side handling of client requests with validation, PostgreSQL storage and instant notifications and cut business response time **from 2 hours to 90 seconds**.
- Integrated **payment systems, web analytics and email delivery**.
- Ran initial performance optimization: added caching, reworked indexes and SQL queries and sped up page response **from 1.8s to 400ms**.

Stack: Python, Django, DRF, PostgreSQL, Nginx, Docker

PROJECTS

ZanityVPN – own VPN service

2022 – Present

Founder and sole engineer

zanityvpn.com

- Grew a private VPN (WireGuard in 2022, then AmneziaWG, Xray-core and 3x-ui with my own config bot) into ZanityVPN, launched in April 2026: **12+ servers in 6 countries**, **200+ active users**.
- Forked the open-source Bedolaga billing bot (Python, PostgreSQL) and web cabinet (React, TypeScript) and **substantially reworked both**: redesigned checkout and subscription management, email support replies with a retry queue, an error log with redelivery from the admin panel.
- **LLM (Claude)** drafts first-line replies to support tickets when confident; CI/CD behind Traefik with health checks, best-server and split routing, per-node iptables firewalls.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

Backend: FastAPI, Django, Django REST Framework, Flask, aiohttp, SQLAlchemy, Alembic, Pydantic, Celery, REST API, gRPC, OpenAPI/Swagger, JWT/OAuth 2.0, asyncio, threading, multiprocessing, pytest, mypy, pyright

ML & AI: PyTorch, MobileNetV3, YOLO, LLM integration (Claude API), Claude Code

Databases: PostgreSQL, MongoDB, Redis, ClickHouse, Elasticsearch, Apache Cassandra

Messaging & Brokers: Apache Kafka, RabbitMQ

DevOps & Infrastructure: Docker, Docker Compose, Kubernetes (Rancher), Nginx, CI/CD (GitHub Actions, GitLab CI), Linux, AWS, S3, WireGuard, Xray-core, iptables

Monitoring & Other: Prometheus, Grafana, Sentry, Git, Agile, microservices

EDUCATION

BSc, Information Security – National Research Nuclear University MEPhI, Moscow 2025

Faculty of Cybernetics and Information Security

LANGUAGES

Russian – Native English – B2+ (Upper-Intermediate), certified (Cambridge FCE)